PARISHRAM 2.0

CLASS 12TH 2026

Electric Charges and Fields

PHYSICS LECTURE-9

BY-DEV AGNANI SIR



Topics to be covered

1. Electric Field due to Uniformly Charged thin Spherical Shell
2. Electric Field due to a Non-Conducting Charged Solid Sphere
3. Electric Field due to a Uniformly Charged Infinite Plane Sheet

Recap

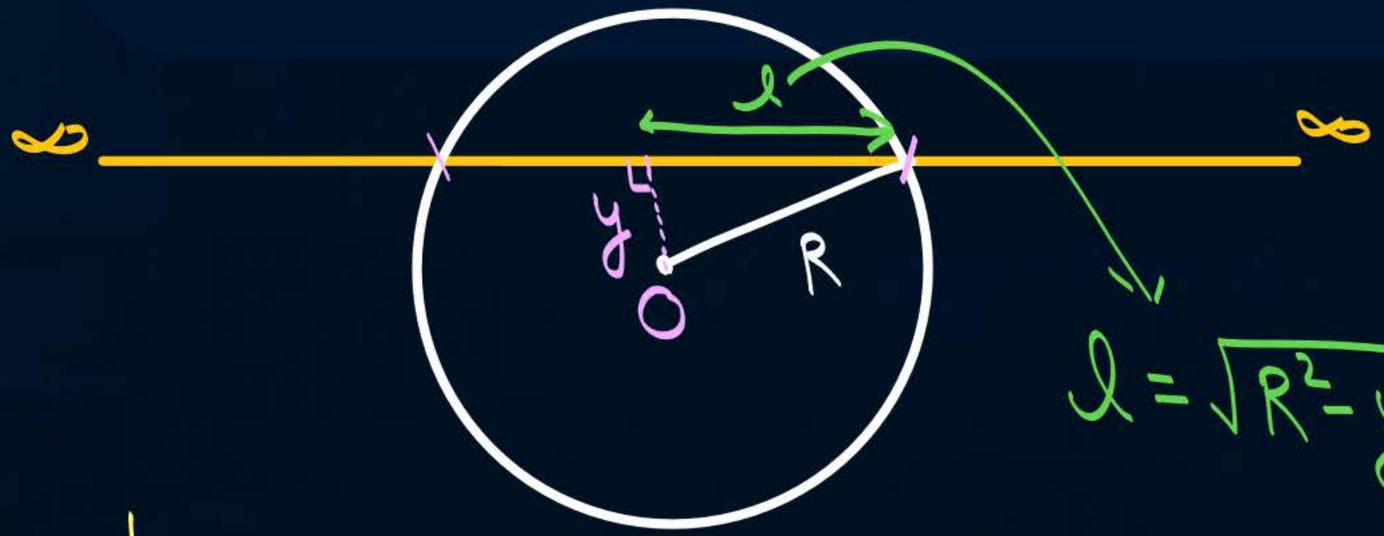
1. Gauss's Law $\phi_E = \oint \vec{E} \cdot \vec{A} = \frac{Q_{enc}}{\epsilon_0}$
2. Electric Field due to a Point charge (By Gauss's Law) $\frac{kQ}{r^2}$.
3. Electric Field due to an infinite linear charge distribution
4.

QUESTION

Good ques:

$\lambda = \frac{Q}{L}$

A **uniformly** charged and infinitely long line having a linear charge **density 'A'** is placed at a normal **distance y** from a **point O**. Consider a sphere of **radius R** with **O** as centre and **R > y**. Electric **flux** through the surface of the sphere is



$l = \sqrt{R^2 - y^2}$

Total length of the wire enclosed in the sphere $= 2l$
$= 2\sqrt{R^2 - y^2}$

Charge enclosed in the sphere $= A\left(2\sqrt{R^2 - y^2}\right)$

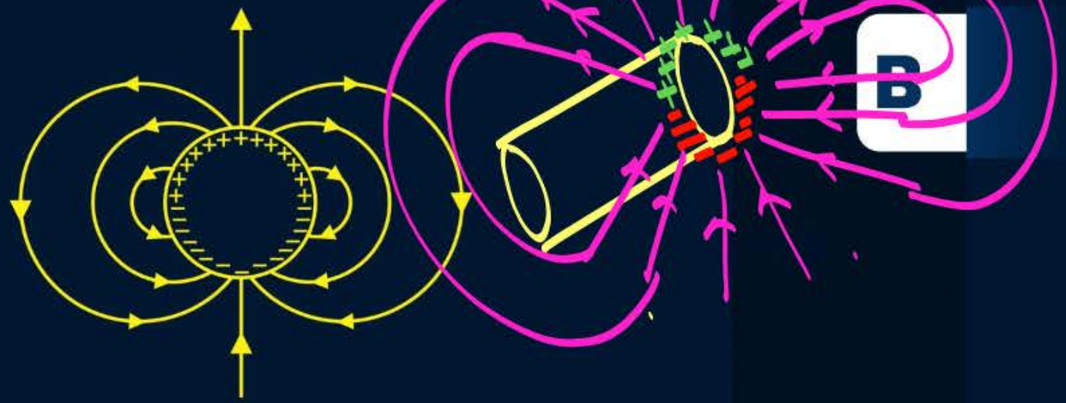
$$\phi_E = \frac{Q_{enc.}}{\epsilon_0} = \boxed{\frac{2A\sqrt{R^2 - y^2}}{\epsilon_0}}$$

QUESTION

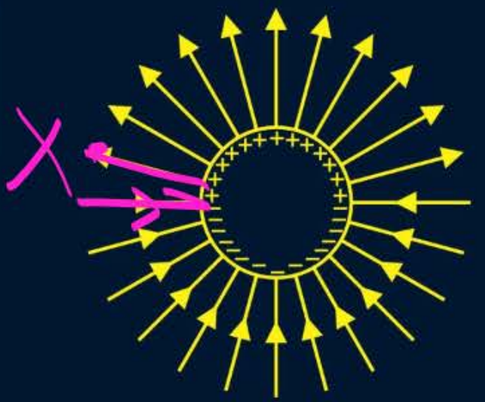
Tough JEE CUET NEET

A long **cylindrical** shell carries **positive** surface **charge** $+\sigma$ in the upper half and **negative** surface **charge** $-\sigma$ in the lower half. The electric field lines around the **cylinder** will look like (Figure are schematic and not drawn to scale).

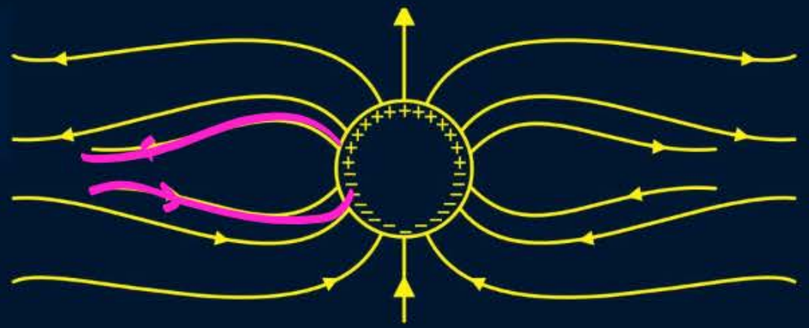
A



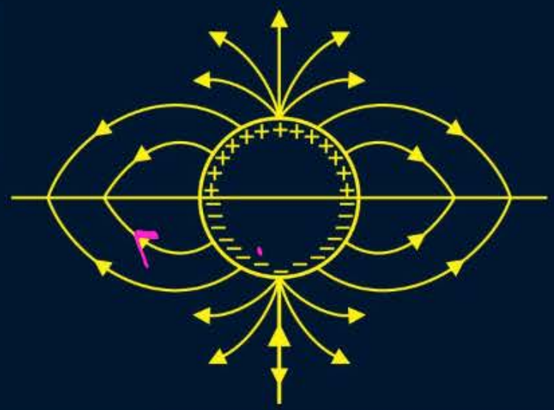
B



C



D



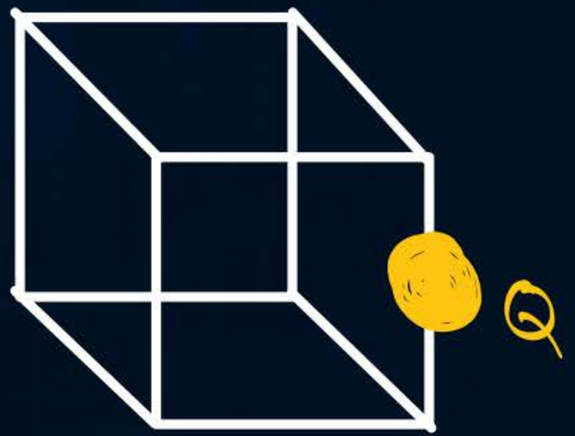
QUESTION

Imagination boosting problem.

A **charge Q** is placed at the centre of the edge of a **cube**. What is the flux through the **cube**?

$Q_{enc} = \frac{Q}{4}$

$\phi_E = ?$

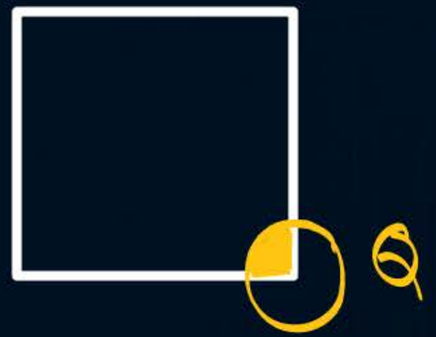


A) Q/ϵ_0

B) ~~$Q/4\epsilon_0$~~

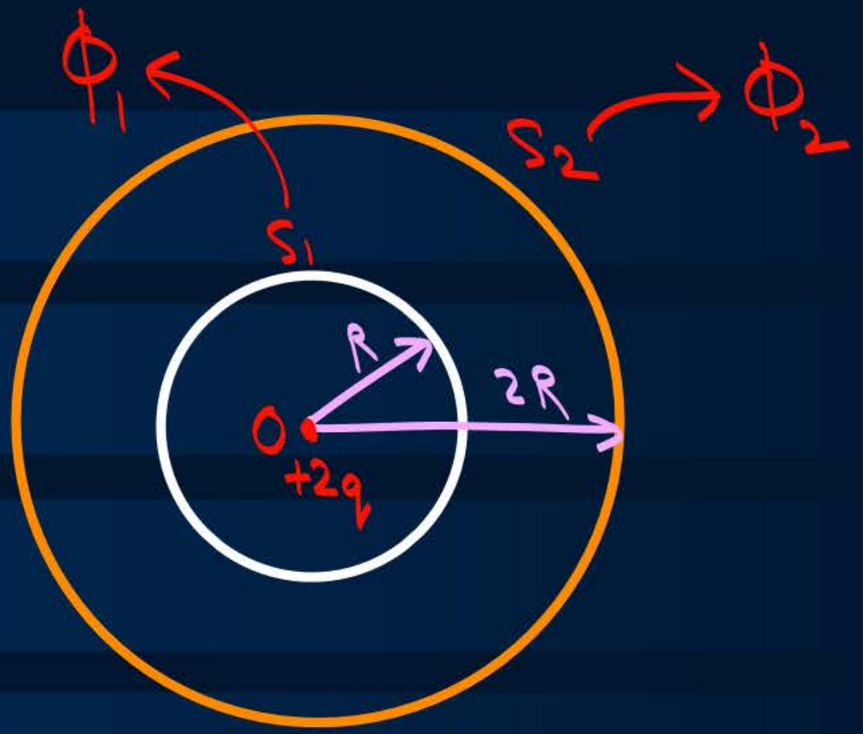
C) $Q/3\epsilon_0$

D) $Q/6\epsilon_0$



QUESTION

Consider two **concentric spherical surface S_1** with **radius R** and **S_2** with **radius 2R** both centered at the origin. There is a charge **+2q** at the origin and there are no other charges. Compare the **flux ϕ_1** through **S_1** with the **flux ϕ_2** through **S_2**.





A $\phi_1 = 2\phi_2$

B $\phi_1 = 4\phi_2$

C $\phi_1 = \phi_2$

D $\phi_1 = \frac{\phi^2}{2}$

QUESTION

$20 \longrightarrow \phi$

$100 \longrightarrow \frac{\phi}{20} \times 100 = 5\phi$

20μC charge is placed **inside** a closed surface, then **flux** linked with the surface is ϕ. If **80μC** charge is put inside the surface then change in **flux** is;



$\Delta\phi = 5\phi - \phi$

$= \boxed{4\phi}$





A 4ϕ

B 5ϕ

C ϕ

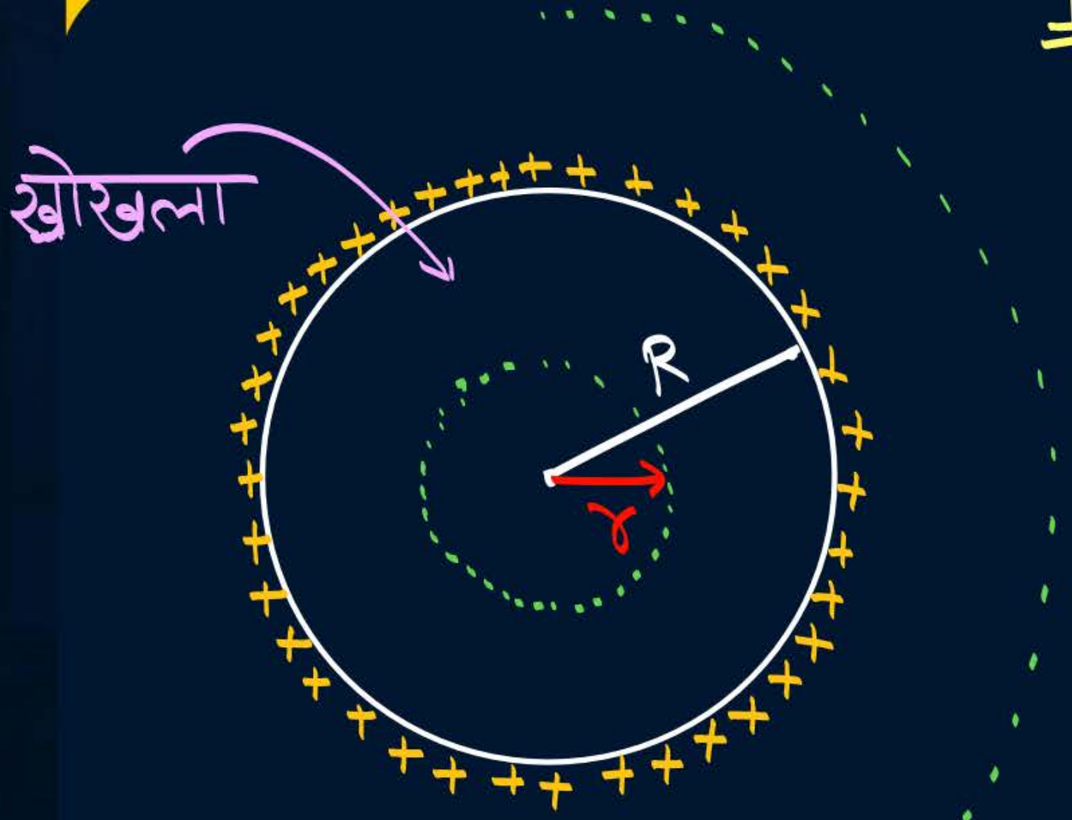
D 8ϕ

When **class-12th board** students are **unable** to attempt **basic conceptual** questions.



Hollow sphere

Electric Field due to Uniformly Charged thin Spherical Shell



inside the Surface :-

$$\oint \vec{E} \cdot \vec{A} = \frac{Q_{enc.}}{\epsilon_0}$$

Since a shell, ∴ No charge present inside.

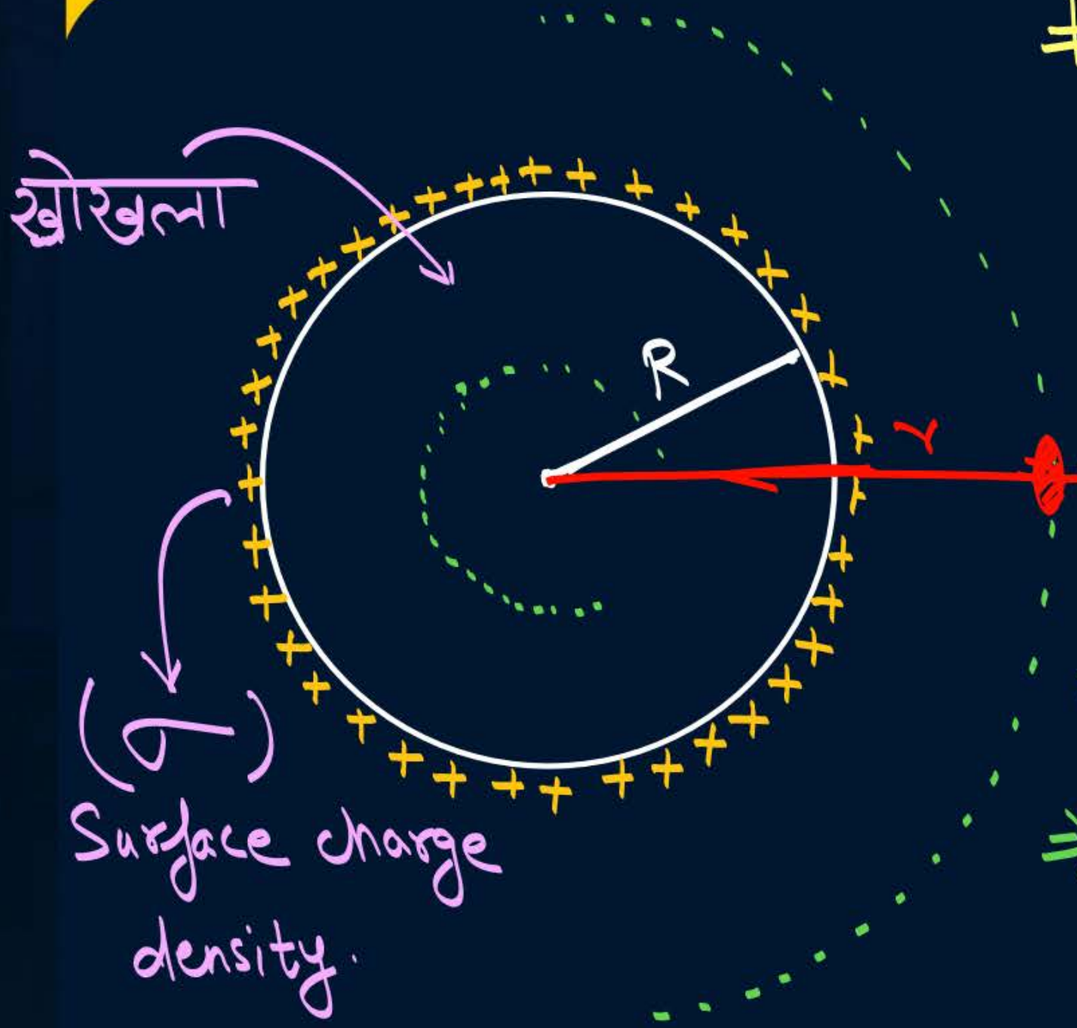
Here $Q_{enc.} = 0$

$$\Rightarrow \boxed{E_{inside} = 0} \quad ; \text{ for } r < R$$

where r is the distance from the centre of the shell and R is the radius of the shell.

Hollow sphere

Electric Field due to Uniformly Charged thin Spherical Shell



Outside the Shell :-

$\gamma > R$

$$\oint \vec{E} \cdot \vec{A} = \frac{Q_{enc.}}{\epsilon_0}$$

$$\Rightarrow \oint E\, dA = \frac{\sigma 4\pi R^2}{\epsilon_0}$$

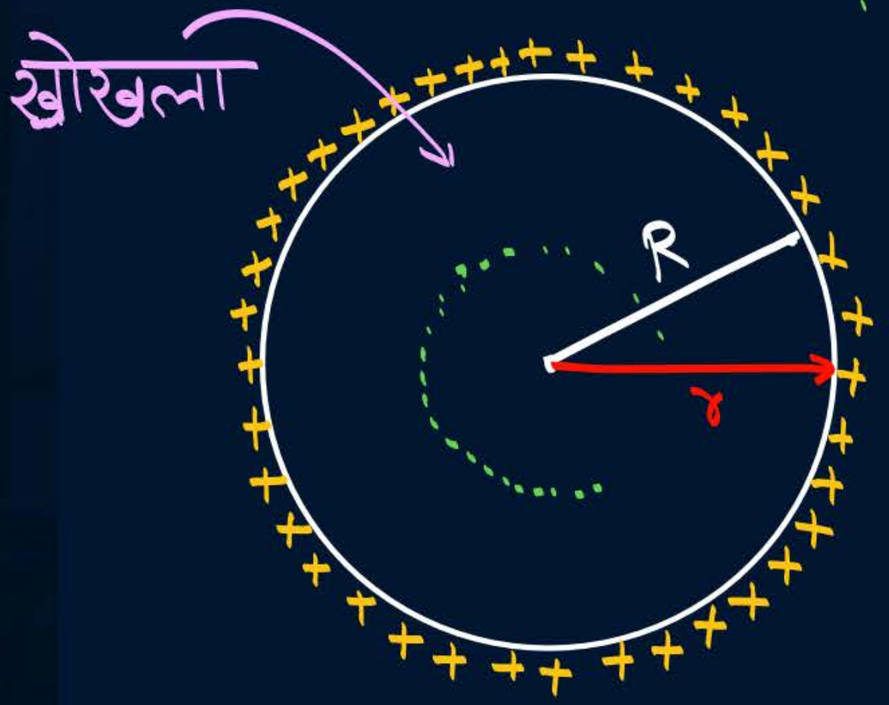
$$\Rightarrow E \oint dA = E(A) = E(4\pi\gamma^2) = \frac{\sigma 4\pi R^2}{\epsilon_0}$$

$$\Rightarrow E(4\pi\gamma^2) = \frac{\sigma 4\pi R^2}{\epsilon_0} \Rightarrow \boxed{E = \frac{\sigma R^2}{\epsilon_0 \gamma^2}} \; ; \gamma > R$$

$$\left[\sigma = \frac{Q}{\text{Area}} = \frac{Q}{4\pi R^2} \Rightarrow Q = \sigma 4\pi R^2\right]$$

Hollow sphere

Electric Field due to Uniformly Charged thin Spherical Shell



On the surface :-

$\Rightarrow r = R$

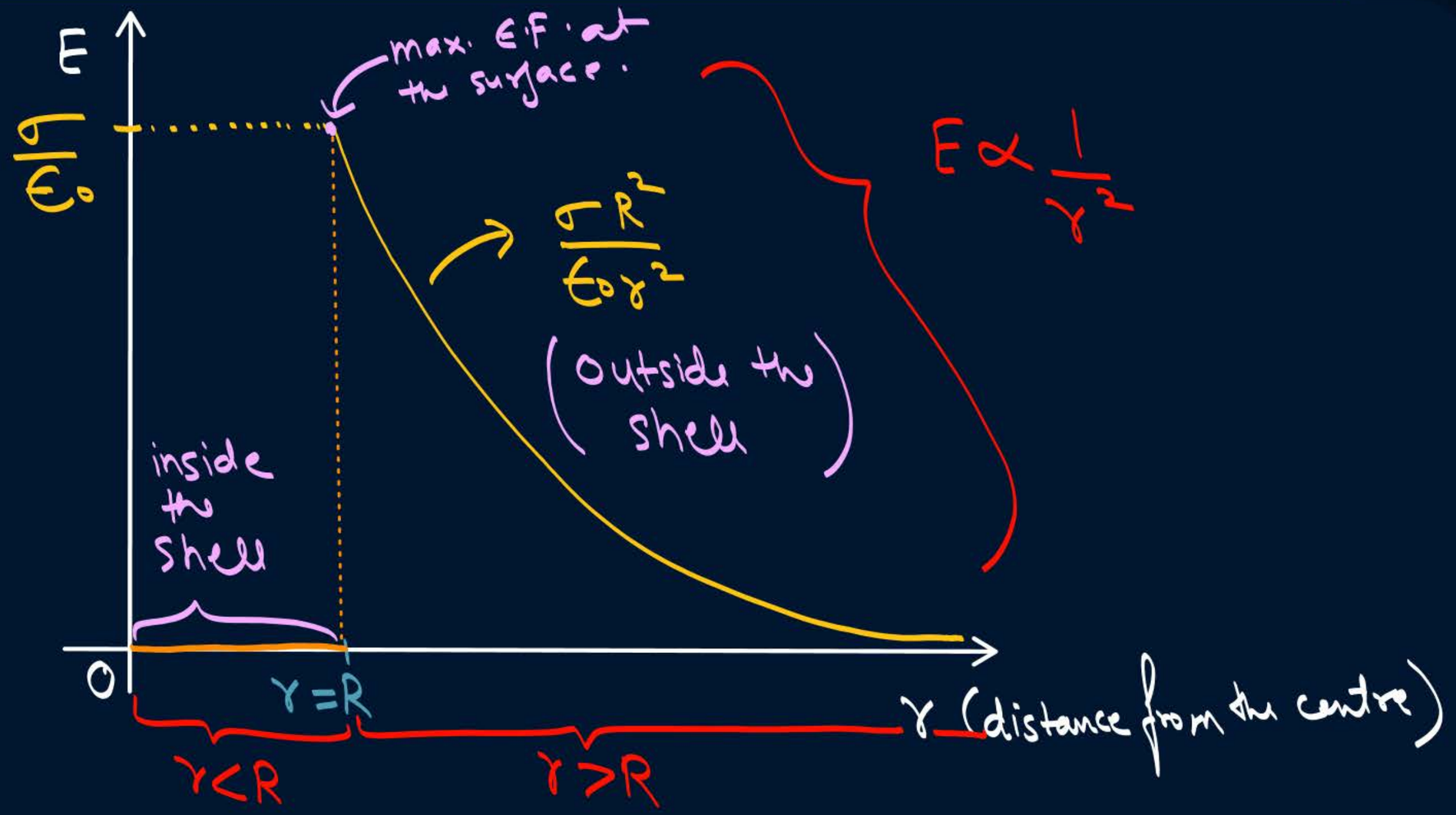
$$E(4\pi R^2) = \frac{\sigma 4\pi R^2}{\epsilon_0}$$

$$\Rightarrow \boxed{E = \frac{\sigma}{\epsilon_0}} \quad ; r = R$$

→ E.F. due to shell on its surface.

E.f. is maximum on the surface of the shell.

Electric Field due to Uniformly Charged thin Spherical Shell



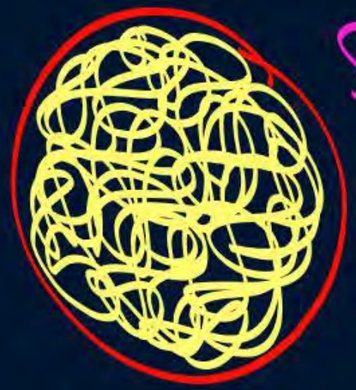
Difference between Conducting and Non-Conducting Solid Spheres

Conducting Solid Sphere	Non-Conducting Solid Sphere
1. Has free electrons.	1. No free electrons.
2. Charges move easily.	2. Charges are fixed in place.
3. E.g. : Metal Ball (Copper, aluminium, brass)	3. E.g. : Plastic or Acrylic Sphere charged by rubbing with wool



Hollow

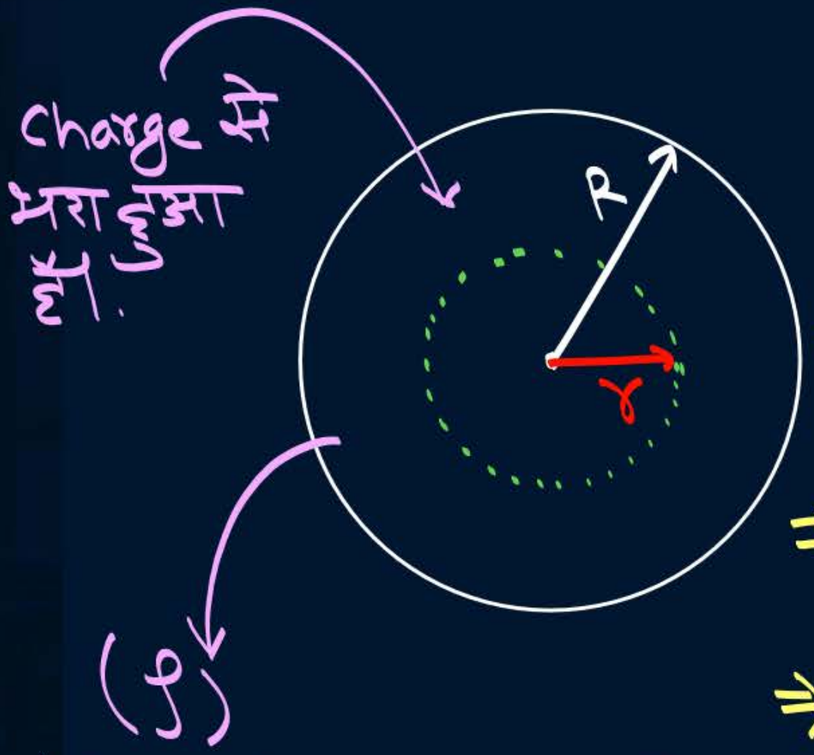




Solid

distance ↑ Stability ↑

Electric Field due to a Non-Conducting Charged Solid Sphere



Inside the sphere :-

$$\oint \vec{E}\cdot\vec{A} = \frac{Q_{enc.}}{\epsilon_0}$$

$$\Rightarrow E\oint dA = \frac{\rho \frac{4}{3}\pi r^3}{\epsilon_0}$$

$$\Rightarrow E(A) = (E)(4\pi r^2) = \frac{\rho 4\pi r^3}{3\epsilon_0}$$

$$\Rightarrow \boxed{E = \frac{\rho r}{3\epsilon_0}} \; ; r < R$$

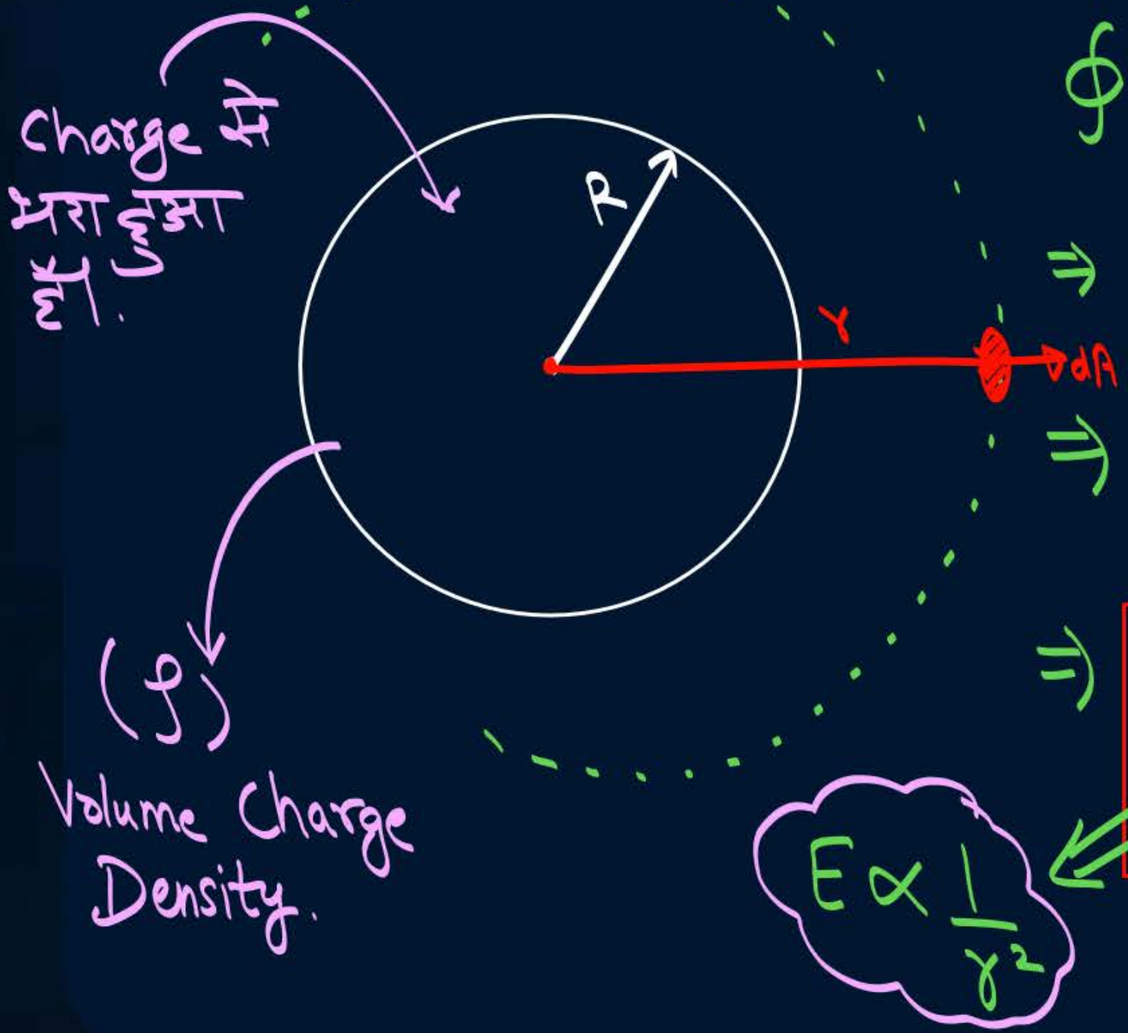
$E \propto r$

$\rho = \frac{Q}{V} \Rightarrow \rho = \frac{Q}{\frac{4}{3}\pi r^3}$

$\Rightarrow Q_{enc.} = \rho \frac{4}{3}\pi r^3$

where r is the distance from the centre of the Solid sphere and R is the Radius of the sphere.

Electric Field due to a Non-Conducting Charged Solid Sphere



Outside the sphere :-

$$\oint \vec{E}\cdot\vec{A} = \frac{Q_{enc.}}{\epsilon_0}$$

$$\Rightarrow E\oint dA = \frac{\rho \frac{4}{3}\pi R^3}{3\epsilon_0}$$

$$\Rightarrow E(4\pi r^2) = \frac{\rho \frac{4}{3}\pi R^3}{3\epsilon_0}$$

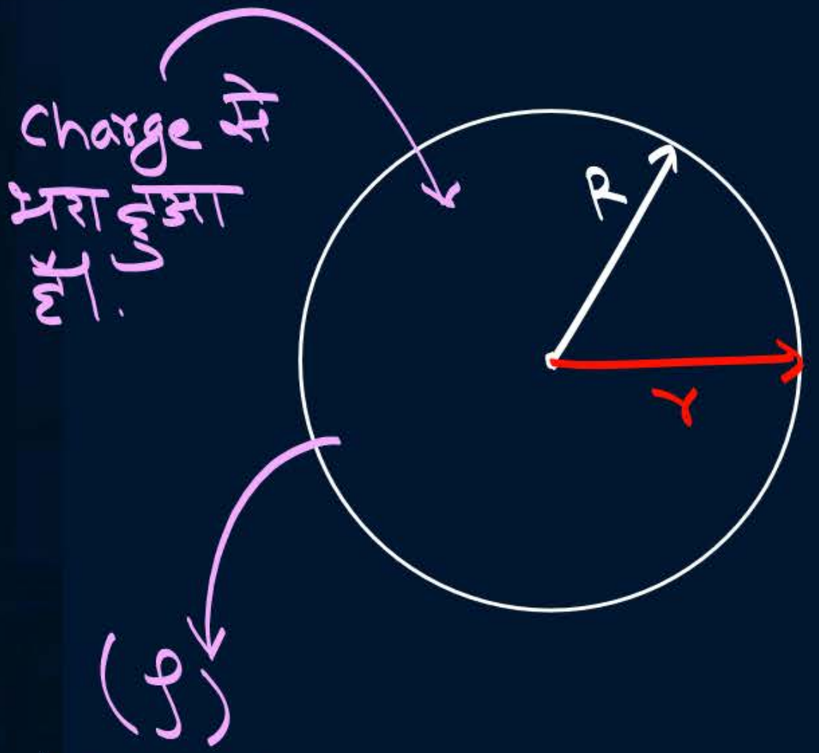
$$\Rightarrow \boxed{E = \frac{\rho R^3}{3\epsilon_0 r^2}} \; ; r > R$$

$$E \propto \frac{1}{r^2}$$

$$\rho = \frac{Q}{\frac{4}{3}\pi R^3}$$

$$\Rightarrow Q_{enc.} = \frac{\rho \frac{4}{3}\pi R^3}{3}$$

Electric Field due to a Non-Conducting Charged Solid Sphere

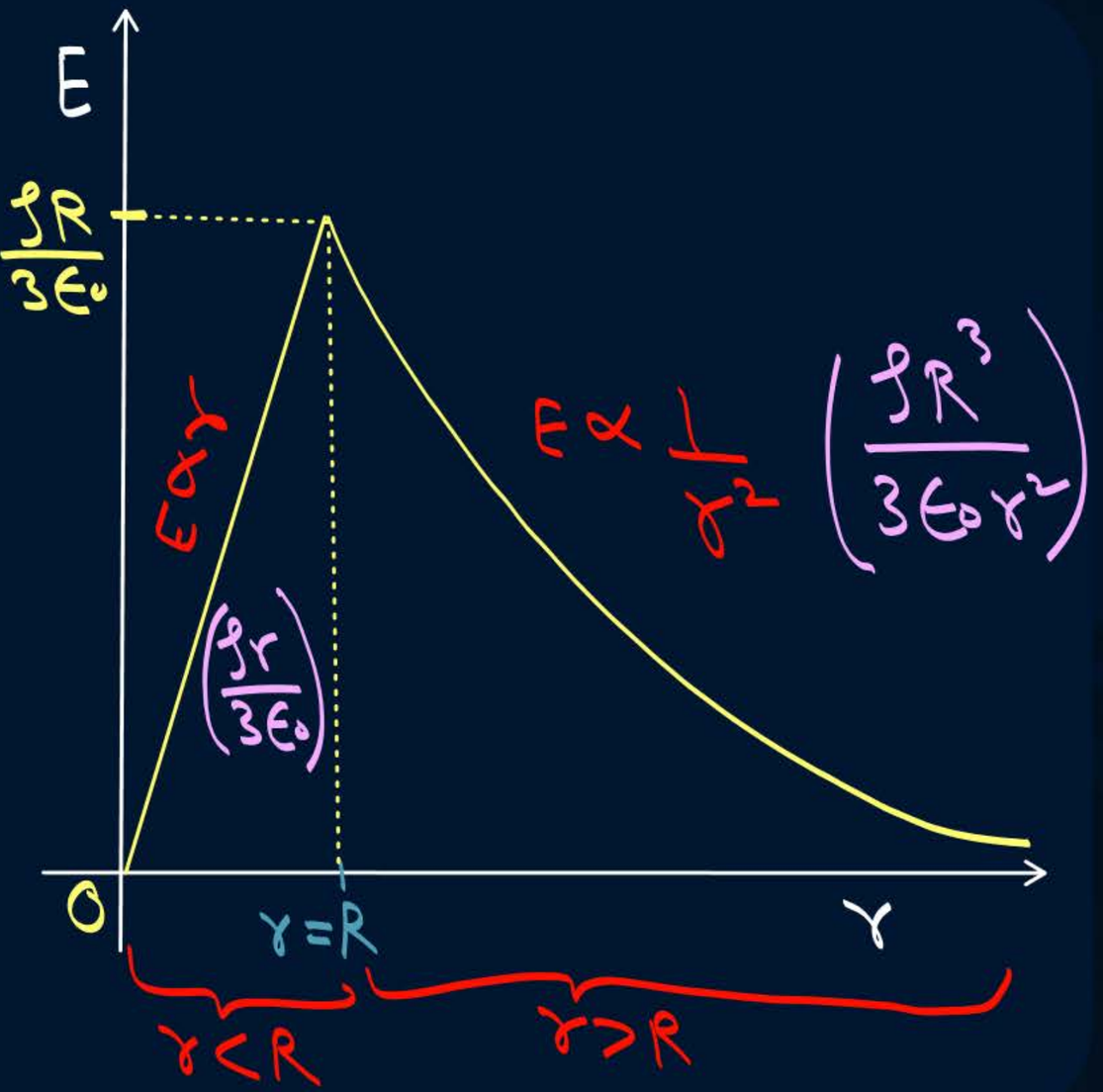


On the surface :-

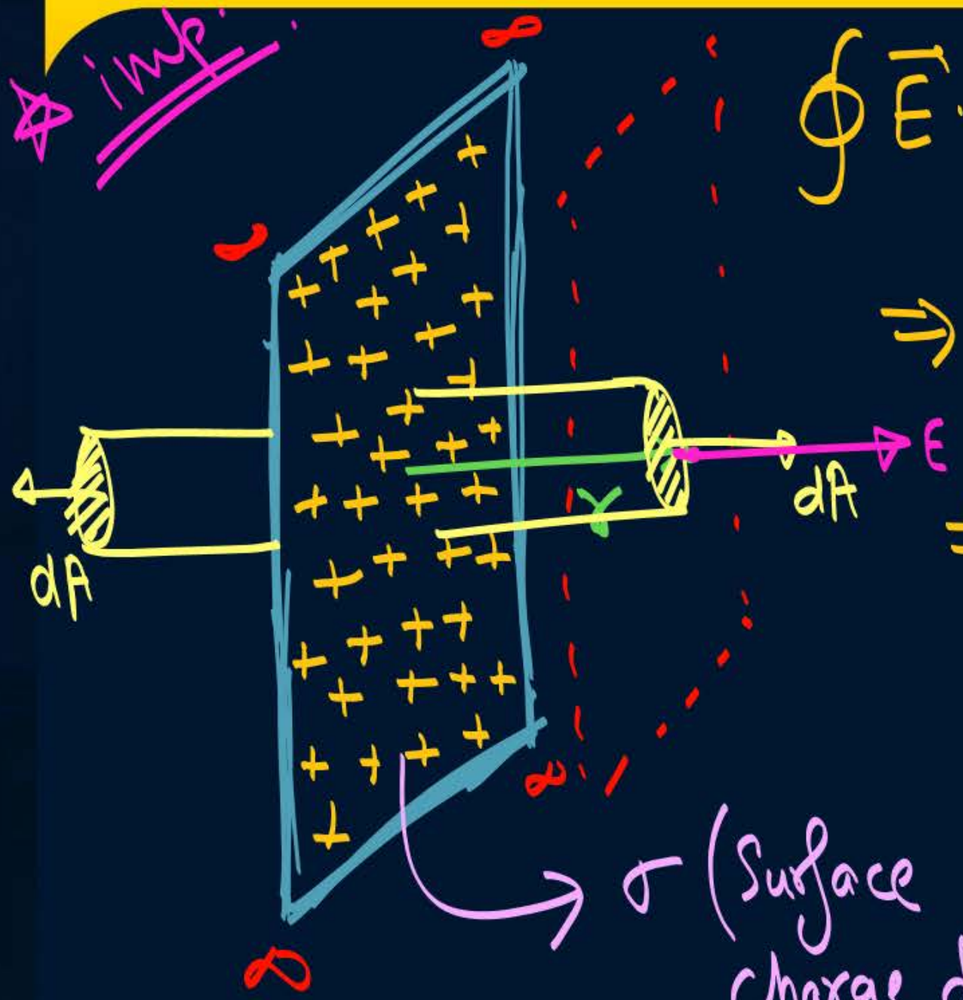
↳ $r = R$

$$E = \frac{\rho R}{3\epsilon_0}$$

↳ Max. E.F.



Electric Field due to a Uniformly Charged Infinite Plane Sheet



imp.

$$\oint \vec{E} \cdot \vec{A} = \frac{Q_{enc.}}{\epsilon_0}$$

$$\Rightarrow E \oint 2\,dA = \frac{\sigma \cdot A}{\epsilon_0}$$

$$\Rightarrow (E)(2A) = \frac{\sigma \cdot A}{\epsilon_0}$$

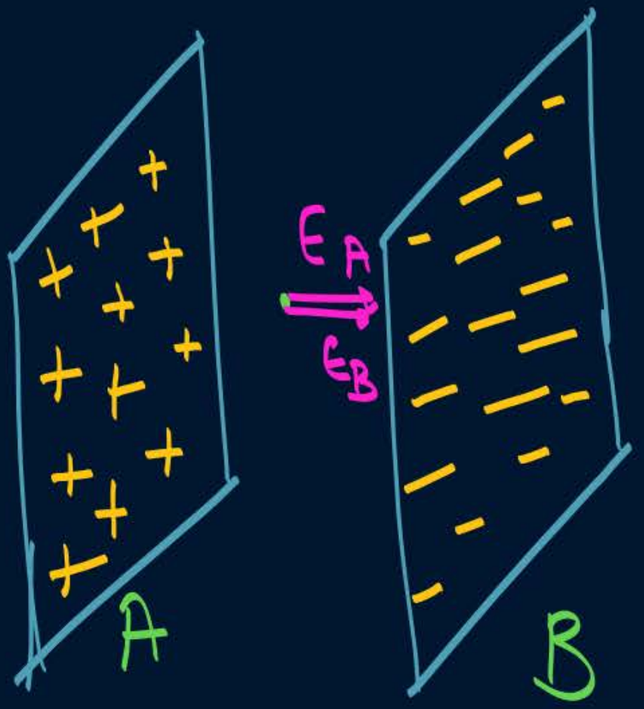
$$\Rightarrow \boxed{E = \frac{\sigma}{2\epsilon_0}}$$

Electric field due to infinite uniformly charged Plane sheet.

$$\sigma = \frac{Q}{\text{Area}}$$

$$\Rightarrow Q_{enc.} = \sigma \cdot A$$

Electric Field due to two Equally and Oppositely Charged Parallel Plane Sheets



$$\text{Net E.F.} = E_A + E_B$$

$$= \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0}$$

$$= \boxed{\frac{\sigma}{\epsilon_0}}$$

← E.F. due to two equally and oppositely charged Parallel Plane sheet at a point between them.

imp.

QUESTION

The electric field at **20 cm** from the centre of a **uniformly charged non-conducting** sphere of **radius 10 cm** is **E**. Then at a distance **5 cm** from the centre it will be;

A 16E

B 4E

C 2E

D Zero



$$E = \frac{\rho R^3}{3\epsilon_0 r^2} \Rightarrow E = \frac{\rho}{3\epsilon_0}\frac{(10)^3}{(20)^2}$$

$$\Rightarrow \frac{\rho}{3\epsilon_0} = \frac{(20)^2 E}{(10)^3}$$

$$\Rightarrow E' = \frac{\rho r}{3\epsilon_0} \Rightarrow E' = \frac{(20)^2 E}{(10)^3} \times (5) = \frac{2 \cancel{10} \cancel{0} \times E \times \cancel{5}}{\cancel{10} \cancel{0} \cancel{0}} = 2E$$



QUESTION

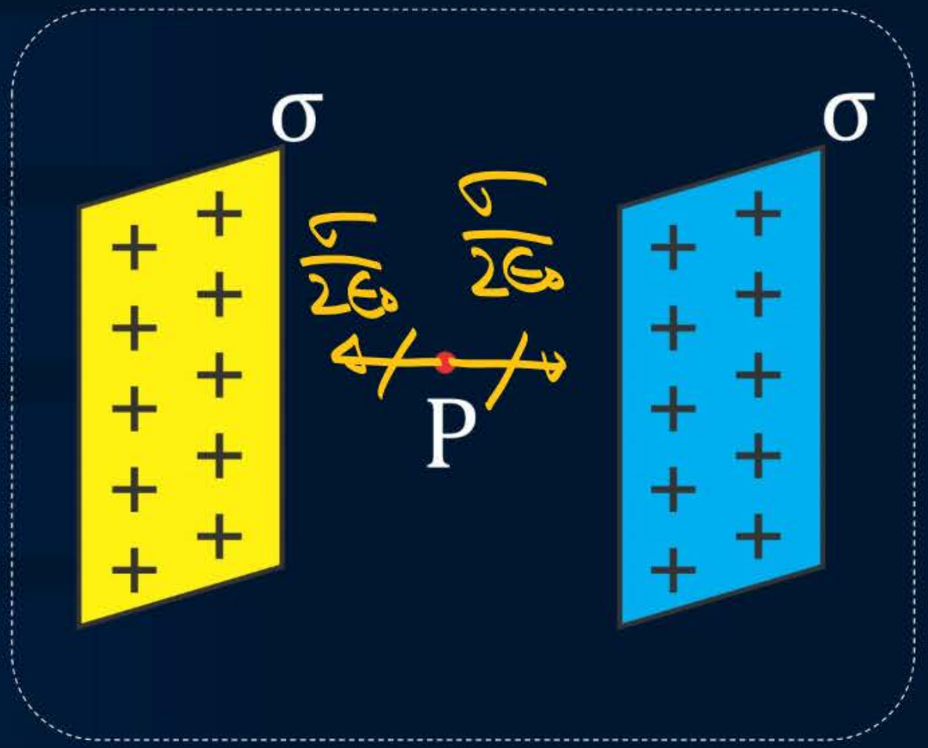
Find **electric field** due to given arrangement of **infinite** plane sheets at **point P.**

A. 0

B. $\frac{\sigma}{\varepsilon_0}$

C. $\frac{\sigma}{2\varepsilon_0}$

D. $\frac{2\sigma}{\varepsilon_0}$



QUESTION

A **hollow metal sphere** of a radius ***R*** is uniformly charged. The **electric field** due to the sphere at a **distance *r*** from the centre:

A increases as r increases for $r < R$ and for $r > R$

B zero as r increases for $r < R$, decreases as r increases for $r > R$

C zero as r increases for $r < R$, increases as r increases for $r > R$

D decreases as r increases for $r < R$ and for $r > R$



Dipole only topic left :

Most imp. Baad ? → Notes Complete ✓

THANK YOU

